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FIRST PRE-BOARD, SESSION- 2024-25

CLASS - XII SUBJECT-PHYSICS THEORY (042)

MAXIMUM MARKS:70 TIME ALLOWED: 3 HRS

General Instructions: -

(1) There are 33 questions in all. All questions are compulsory.
(2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
(3) All the sections are compulsory.
(4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study based questions of four marks each and Section E contains three long answer questions of five marks each.
(5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
(6) Use of calculators is not allowed.
(7) You may use the following values of physical constants where ever necessary
i) $c = 3 \times 10^8$ m/s ii) $m_e = 9.1 \times 10^{-31}$ kg iii) $e = 1.6 \times 10^{-19}$ C
iv) $\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1}v) $h = 6.63 \times 10^{-34}$Js vi) $\varepsilon_0 = 8.854 \times 10^{-12} C^2N^{-1}m^{-2}$
vii) Avogadro's number = 6.023×10^{23} per gram mole

Section- A

1:- Beams of electrons and protons move parallel to each other in the same direction. They

(a) attract each other

(b) repel each other

(c) neither attract not repel

(d) force of attraction or repulsion depends upon speed of beams

2:- If a diamagnetic substance is brought near the N-pole or the S-pole of a bar magnet, then it is

(a) attracted by the both poles

(b) repelled by both the poles

(c) repelled by the N-pole and attracted by the S-pole

(d) attracted by the N-pole and repelled by the S-pole

3:- An isolated point charge particle produces an electric field E at a point 3m away from it. The distance of the point at which the field is $\frac{E}{4}$ will be

(a) 2 m (b) 3 m (c) 4 m (d) 6 m

4:- In an n-type silicon, which of the following statement is correct?

(a) Electrons are majority charge carriers and trivalent atoms are the dopants.

(b) Electrons are minority charge carriers and pentavalent atoms are the dopants.

(c) Holes are minority charge carriers and pentavalent atoms are the dopants.

(d) Holes are majority charge carriers and trivalent atoms are the dopants.

5:- What is the work done in moving a 5μC point charge from corner A to corner at B of a square ABCD, when a 20μC exists at the center of square?

(a) 100 J (b) 25 J (c) 0 J (d) 5 J

6:- The radius of the innermost electron orbit of a hydrogen atom is 5.3×10^{-11}m. The radius of the second excited orbit is

(a) 1.01×10^{-10} m (b) 1.59×10^{-10} m

(c) 2.12×10^{-10} m (d) 4.77×10^{-10} m

7:- Frequency of wave is 6×10^{10} Hz. The wave is

(a) Radio wave (b) Micro wave

(c) X-Ray (d) None of these

8:- The current in the primary coil of a pair of coils changes from 7A to 3A in 0.04 s. The mutual inductance between the two coils is 0.5H. The induced emf in the secondary coil is.

(a) 50 V (b) 75 V (c) 100 V (d) 220 V

9:- If an alternating voltage is represented as E=141 Sin 628t, then the rms value of the voltage and the frequency are respectively

(a) 14IV, 628Hz (b) 100V, 50Hz

(c) 100V, 100Hz (d) 14IV, 100Hz

10:- A current carrying wire kept in a uniform magnetic field will experience a maximum force, when it is

(a) Perpendicular to the magnetic field

(b) Parallel to the magnetic field

(c) At an angle of 45° to the magnetic field

(d) at an angle of 60° to the magnetic field

11:- Two convex and conclave lenses are in contact and having focal length 12cm and 18cm, respectively. Focal length of joint lens will be

(a) 50 cm (b) 45 cm (c)36 cm (d) 18 cm

12:- The de-Broglie wave of a moving particle does not depend on.

(a) Mass (b) Charge (c) Velocity (d) Momentum

For questions 13 to 16 , two statements are given- one labelled Assesrtion (A) and other labelled Reason (R). Select the correct answers to these questions from the options as given below-

a) If both assertion and reason are true and reason is correct explanation of assertion.

b)) If both assertion and reason are true and reason is not the correct explanation of assertion.

c) If assertion is true but reason is false.

d) If both assertion and reason are false

13:- Assertion (A):- Charge on all the condensers connected in series is the same.

Reason (R):- Capacitance of capacitor is directly proportional to charge on it.

14:- Assertion (A):- The pole of magnet cannot be separated by breaking into two pieces.

Reason(R):- The magnetic moment will be reduced to half when a magnet is broken into the equal pieces.

15:- Assertion (A):- Photoelectric current depends on the intensity of incident light.

Reason(R):- Number of photoelectric emitted per second is directly proportional to the intensity of incident radiation.

16:- Assertion (A):- An uncontrolled chain reaction can become highly explosive.

Reason(R):- In uncontrolled chain reaction neutrons released in fission reaction are again and again absorbed by the fissile isotopes.

Section- B

17:- A plane electromagnetic wave of frequency 25MHz travels in free space along the x- direction. At a particular point in space and time, $\vec{E}$ =6.3$\hat{j}$ Vm^{-1}. What is $\vec{B}$ at this point?

18:- What do you mean by process of rectification. Draw a circuit diagram of full wave rectifier.

19:- The work function of a certain metal is 4.2eV. Will this metal give photoelectric emission for incident radiation of wave length 330nm?

Or

Sketch the graph showing (i) variation of stopping potential with frequency of incident radiation

(ii) Variation of photoelectric current with anode potential at constant intensity and different frequency.

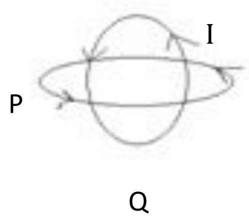
20:- Using Huygens principle show that, for a parallel beam incident on a reflection surface, the angle of reflection is equal to the angle of incidence.

21:- A battery of emf E, and internal resistance r, gives a current of 0.5A with an external resistance of 12 Ohm and a current of 0.25A with an external resistance of 25 Ohm. Calculate (i) internal resistance of the cell and (ii) emf of the cell.

Section- C

22:- (i) Write Biot- Savart's law for magnetic field at a point due to a small current element in vector form.

(ii) Two identical circular wires P and Q each of radius R and carrying current I are kept in perpendicular planes such that they have a common centre as shown in figure.



Find the magnitude and direction of net magnetic field at the common centre of two coils.

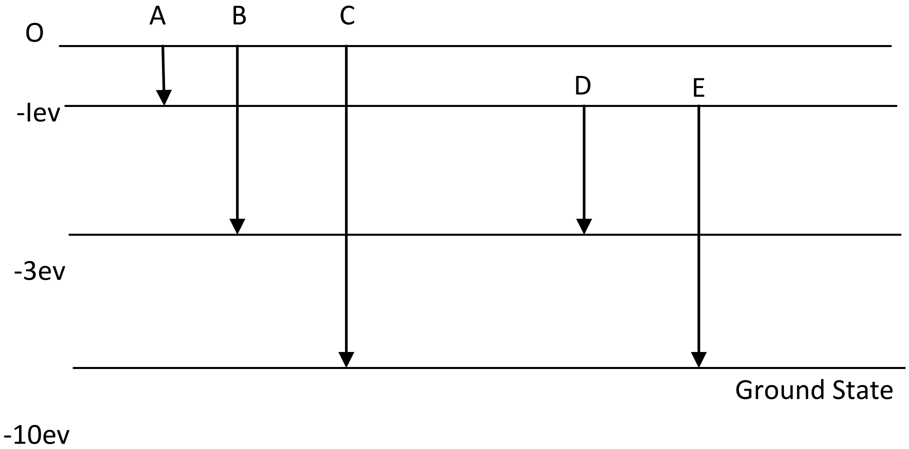
23:- (i) Draw a graph to show the variation of the angle of deviation with that of the angle of incidence for a prism.

(ii) A ray of light passes through an equilateral glass prism, such that the angle of incidence is equal to the angle of emergence. If the angle of emergence us ¾ times the angle of the prism, calculate the refractive index of the glass prism. Also find speed of light in prism.

24:- State Gauss's theorem in electrostatics. Using this theorem, derive an expression for the electric field intensity due to an infinitely long, straight wire of linear charge density λ.

25:- Define the term drift velocity and relaxation time.Derive an expression for drift velocity of free electrons in a conductor in terms of relaxation time.

26:- The energy levels of an atom of element X are shown in figure.



Which one the level transmissions with result in the emission of photons of wavelength 620 nm?

Support your answer with mathematical calculations.

27:- Calculate the binding energy per nucleon of $^{35}_{17}$Cl nucleus. Given that

Mass of $^{35}_{17}$Cl = 34.980000 u

Mass of proton= 1.007825 u

Mass of neutron= 1.008665 u

1 u = 931mev.

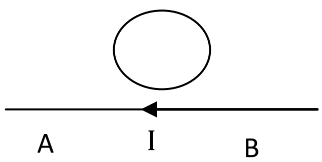
Or

(i) Define Binding energy of a nucleus.

(ii) Draw a plot showing the variation of Binding energy per nucleon with mass number.

28:- State (i) Faraday's law (ii) Lenz's law, of electromagnetic induction.

The electric current flowing in a wire in the direction B to A is decreasing, what is the direction of the magnetic field in the metallic loop kept above the wire shown is figure.



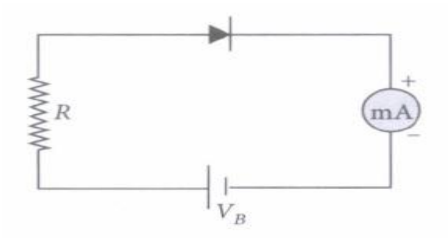
Section- D

(Case Study Based Questions)

Read the following paragraph and answer the question follows:

29:- The potential barrier in the p-n junction diode is the barrier in which the charge requires additional force for crossing the region. In other words, the barrier in which the charge carrier stopped by the obstructive force is known as the potential barrier. When a p-type semiconductor is brought into a close contact with n-type semiconductor, we get a p-n junction with a barrier potential 0.4 V and width of depletion region is

4.0×10^{-7} m. This p-n junction is forward biased with a battery of voltage 3V and negligible internal resistance, in series with a resistor of resistance R, ideal milliammetre and key K as shown in figure. When key is pressed, a current of 20 mA passes through the diode.



(i) The intensity of the electric field in the depletion region when p-n junction is unbiased is-

(a) 0.5×10^6 V m^{-1} (b) 1.0×10^6 V m^{-1} (c) 2.0×10^6 V m^{-1} d) 1.5×10^6 V m^{-1}

(ii) The resistance of resistor R is-

(a) 150 Ω (b) 300 Ω (c) I30Ω (d) 180 Ω

(iii) In a p-n junction, the potential barrier is due to the charges on either side of the junction, these charges are-

(a) majority carriers (b) minority carriers

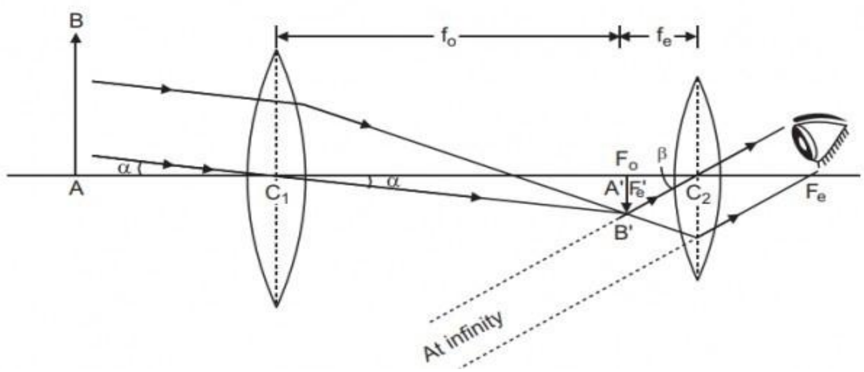
(c) both (a) and (b) (d) fixed donor and acceptor ions.

(iv) If the voltage of the potential barrier is V_0. A voltage V is applied to the input, at what moment will the barrier disappear?

(a) $V< v_0$ (b) $V = V_0$ (c) $V > V_0$ (d) $V<< v_0$

30:- Astronomical telescope

An astronomical telescope is an optical instrument which is used for observing distinct images of heavenly bodies like stars, planets etc. It consists of two lenses. The lens having large aperture and greater focal length is used as objective lens and the lens having smaller aperture and smaller focal length is used as eye piece. In normal adjustment of telescope, the final image is formed at infinity. Magnifying power of astronomical telescope in normal adjustment is defined as the ratio of the visual angle subtended by final image to the visual angle subtended by the object at least distance of distinct vision, and it is given by formula: m = $\frac{f_o}{f_e}$.



i) An astronomical telescope of magnifying power 7 consists of two thin lenses 40cm apart, in normal

adjustment The focal length of the lenses are-

(a) 5cm,35cm (b) 7cm,35cm (c)17cm, 35cm (d) 5cm, 30cm

ii) An astronomical telescope has a angular magnification 10. In normal adjustment, distance between the objective and eye piece is 22cm. the focal length of the objective lens is –

(a)25cm (b)10cm (c)15cm (d)20cm.

iii) For large magnifying power of astronomical telescope-

(a) $f_0<<f_e$ (b) $f_0=f_e$ (c) $f_0>>f_e$ (d)none of these

iv) How does the magnifying power of a telescope change on increasing the diameter of its objective?

a) Independent b) Doubled c) Halved d) Becomes zero

OR

How much intensity of the image is increased if the diameter of the objective of a telescope is doubled?

a) Two times b) Four times c) Eight times d) Sixteen times

Section- E

31:- (i) In a young's double slit experiment, find the condition for constructive and destructive interference. Hence, write the expression for the distance between two consecutive bright or dark fringe.

(ii) The fringe width in a Young's double slit interference pattern is $2.4\text{x}10^{-4}$m, when red light of wavelength 4000 A° is used.

Or

(i) What is meant by diffraction of light? What are the conditions for the angular width of secondary? Maxima and secondary minima obtained in diffraction pattern.

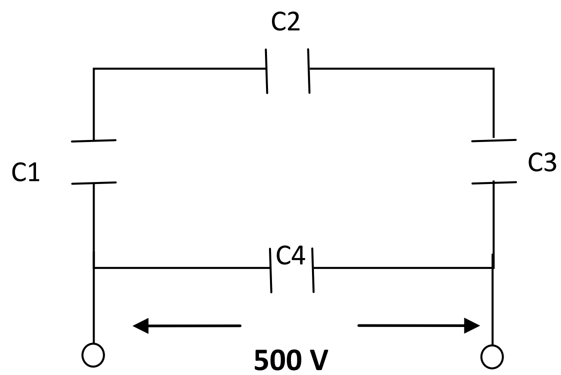
(ii) Determine the angular separation between central maximum and first order maximum of the diffraction pattern due to a single slit of width 0.25 mm when light of wavelength 5890 A^0 is incident on it normally.

32:- (i) Obtain an expression for the capacitance of a parallel plate (air) capacitor.

(ii) A network of four capacitors each of 12 μF capacitance is connected to 500V supply as shown in the figure.

Determine (a) Equivalent capacitance of the network.

(b) Charge on each capacitor.



Or

(i) what in equipotential surface?

(ii) Sketch equipotential surface for

(a) a positive point charge

(b) Two equal and opposite charges separated by a small distance.

(iii) An infinite plane sheet of charge density 10^{-8} Cm^{-2} is held in air. In this situation how far apart are two equivalent surfaces, whose potential difference is 5V.

33:- (i) A device X is connected across an a.c source of voltage $E = E_o \sin \omega t$ The current through X is given as $I = I_o \sin(\omega t + \pi/2)$. Identify the device X. Write the expression for its reactance. Draw graph showing variation of voltage and current with time over one cycle of a.c for X.

(ii) A series LCR circuit consists of a resistance of 10 Ω, a capacitor of reactance 60 Ω and an inductor coil. The circuit is found to resonate when put across 300V, 100 Hz supply. Calculate (i) the inductance of the coil (ii) current in the circuit at resonance.

Or

With the help of a labeled diagram, explain the function, principle, Construction and working of an a.c generator. Derive expression for induced emf.

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